Experiment-10

Develop an application to change the font and color of the text and display toast message when the user presses the button**.**

AIM:

To develop an Android application that allows the user to change the font and color of text displayed on the screen and display a Toast message when a button is pressed.

ALGORITHM:

1. **Initialize the Android Application:**
   - Set up a new Android project in **Android Studio**.
   - Create a **MainActivity** Java file for the main screen of the application.
2. **Design the User Interface (UI):**
   - In the **XML layout file**, add the following UI components:
     - **TextView**: For displaying the text whose font and color will be changed.
     - **Button**: To trigger the change in font and color, and show a **Toast message**.
     - **Spinner or Color Picker**: To let the user choose a color (optional, or a simple button for a predefined color change).
     - **RadioButtons** or a **Dropdown Menu** to select different font styles (optional).
3. **Declare Variables:**
   - Create references for the **TextView**, **Button**, **Color Picker** (if used), and **Radio Buttons** (if used to change font style).
   - Define variables for the **font style**, **color**, and the **message** to display in the **Toast**.
4. **Set Event Listener for the Button:**
   - Set an **OnClickListener** for the **Button**.
   - Inside the onClick() method, write the logic to:
     - Change the **font** of the **TextView** based on the selected font style.
     - Change the **color** of the **TextView** based on the selected color.
     - Display a **Toast message** indicating the action has been performed.

5. **Implement Font Style Changes:**
    - Change the **TextView's font** using setTypeface() to apply a selected font style (e.g., bold, italic).
    - You can either use predefined styles (like Typeface.BOLD or Typeface.ITALIC) or let the user choose from a list of font styles.
6. **Implement Color Change:**
    - Change the **TextView's text color** by using the setTextColor() method and applying the selected color (e.g., Color.RED, Color.BLUE, or a color from the **Color Picker**).
    - If using a color picker or dropdown, get the selected color and set it on the **TextView**.
7. **Show Toast Message:**
    - Display a **Toast** message to the user when the button is clicked, informing them that the text style or color has changed. You can also show a message like "Font and color changed!" or any other appropriate message.
8. **End the Operation:**
    - After the changes are made, the application will display the updated **TextView** with the new font and color, and the **Toast message** will appear to provide feedback.

CODE:

activity_main.xml:

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    android:id="@+id/main"
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        android:textSize="24sp"
        android:textColor="#000000"
        android:padding="16dp"/>

    <Button
        android:id="@+id/btnColor"
```

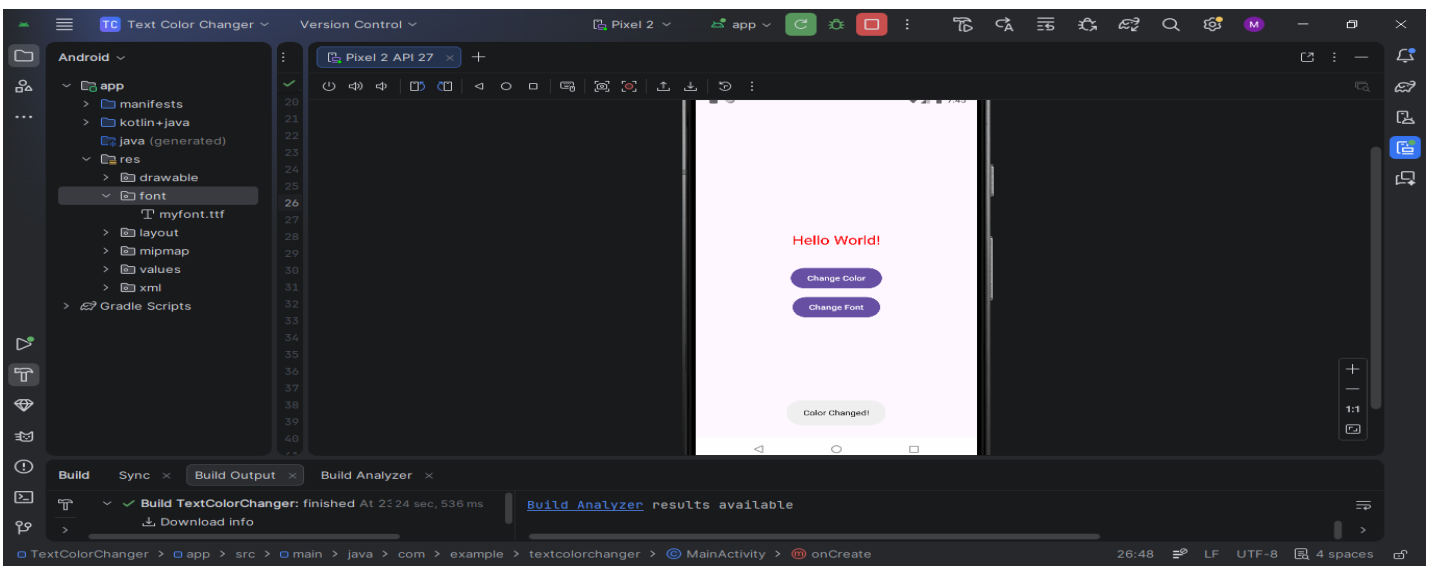
```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Change Color"
        android:layout_marginTop="20dp"/>

    <Button
        android:id="@+id/btnFont"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Change Font"
        android:layout_marginTop="10dp"/>

</LinearLayout>
```

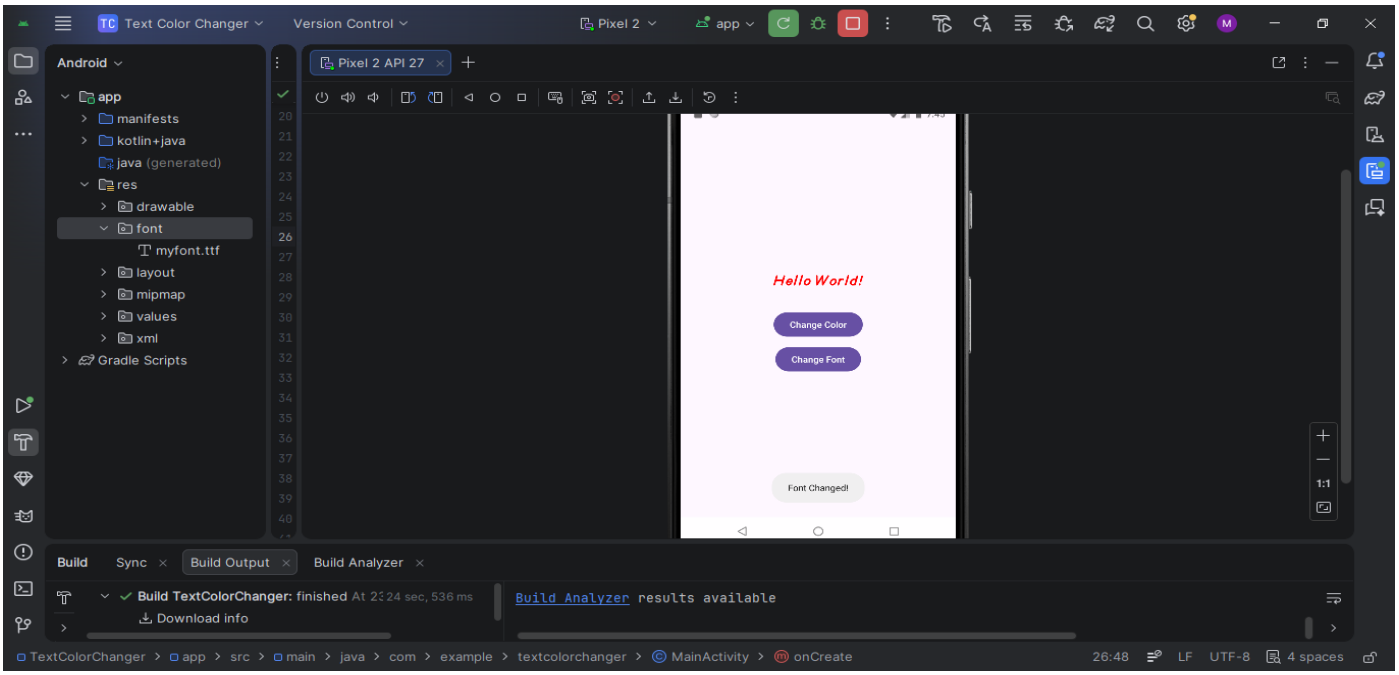
MainActivity.java:

```
package com.example.textcolorchanger;

import android.graphics.Color;
import android.graphics.Typeface;
import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.content.res.ResourcesCompat;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    TextView textView;
    Button btnColor, btnFont;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        // Edge-to-edge support
        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
            Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
            return insets;
        });

        // Initialize views
        textView = findViewById(R.id.textView);
        btnColor = findViewById(R.id.btnColor);
        btnFont = findViewById(R.id.btnFont);
```

```java
        // Change Color Button
        btnColor.setOnClickListener(v -> {
            textView.setTextColor(Color.RED);
            Toast.makeText(this, "Color Changed!", Toast.LENGTH_SHORT).show();
        });

        // Change Font Button
        btnFont.setOnClickListener(v -> {
            try {
                Typeface typeface = ResourcesCompat.getFont(this, R.font.myfont);
                if (typeface != null) {
                    textView.setTypeface(typeface);
                    Toast.makeText(this, "Font Changed!", Toast.LENGTH_SHORT).show();
                } else {
                    Toast.makeText(this, "Font not found!", Toast.LENGTH_SHORT).show();
                }
            } catch (Exception e) {
                Toast.makeText(this, "Error loading font", Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

Output:





RESULT:

An mobile application is developed in Android studio to change the font and color of the text and display toast message when the user presses the button.